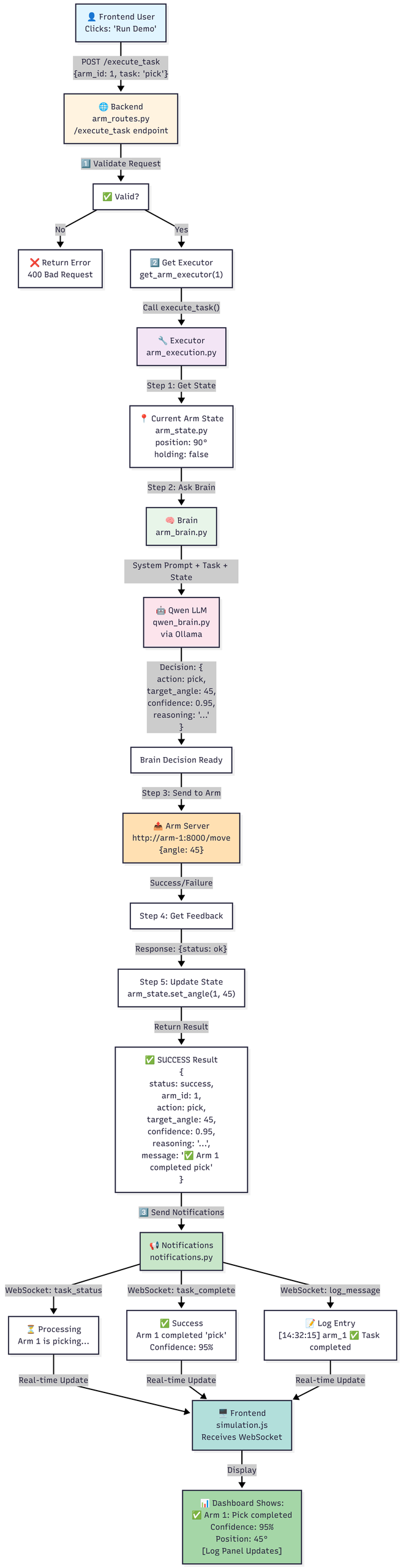

👤 Frontend User Clicks: 'Run Demo'
POST /execute_task {arm_id: 1, task: 'pick'}
🌐 Backend arm_routes.py /execute_task endpoint
1️⃣ Validate Request
✅ Valid?
No
Yes
❌ Return Error 400 Bad Request
2️⃣ Get Executor get_arm_executor(1)
Call execute_task()
🔧 Executor arm_execution.py
Step 1: Get State
📍 Current Arm State arm_state.py position: 90° holding: false
Step 2: Ask Brain
🧠 Brain arm_brain.py
System Prompt + Task + State
🤖 Qwen LLM qwen_brain.py via Ollama
Decision: { action: pick, target_angle: 45, confidence: 0.95, reasoning: '...' }
Brain Decision Ready
Step 3: Send to Arm
📤 Arm Server http://arm-1:8000/move {angle: 45}
Success/Failure
Step 4: Get Feedback
Response: {status: ok}
Step 5: Update State arm_state.set_angle(1, 45)
Return Result
✅ SUCCESS Result { status: success, arm_id: 1, action: pick, target_angle: 45, confidence: 0.95, reasoning: '...', message: '✅ Arm 1 completed pick' }
3️⃣ Send Notifications
📢 Notifications notifications.py
WebSocket: task_status
WebSocket: task_complete
WebSocket: log_message
⏳ Processing Arm 1 is picking...
✅ Success Arm 1 completed 'pick' Confidence: 95%
📝 Log Entry [14:32:15] arm_1 ✅ Task completed
Real-time Update
Real-time Update
Real-time Update
🖥️ Frontend simulation.js Receives WebSocket
Display
📊 Dashboard Shows: ✅ Arm 1: Pick completed Confidence: 95% Position: 45° [Log Panel Updates]